Outputs Explanation

The document goes through every output file produced by the programs in the source code archive and explains what is inside it, how to read it, and what it told us. Figures are described but not shown here; the files themselves sit in the numbered output folders.

differential_expression_analysis.py

This is the main differential expression script. It answers Q1 (the genotype question, mutant vs wild-type at 18 months), Q2 (the ageing question, old vs young wild-type) and Q3 (the overlap between the two), so it produces the largest set of outputs. Throughout, a gene counts as significant if its q-value is below 0.05 and its absolute log2 fold change is above 1, which means the gene has at least doubled or halved.

- **q1_genotype_DEG_results.csv:** the full Q1 results, one row for every gene tested. The columns are the gene id and name, the mean expression, the log2 fold change (positive means higher in the mutant, negative means lower), the standard error, the test statistic, the raw p-value, and the q-value after multiple testing correction. Applying the significance cut-off gives 5,541 genes, of which 4,252 go down in the mutant and 1,289 go up, so the dominant direction is switching off. This is the table almost every later script feeds on.
- **q2_age_DEG_results.csv:** exactly the same columns but for the ageing comparison in wild-type fish only (37 months against 11 months). Here a positive fold change means the gene goes up as the fish get older. It gives 515 significant genes, so ageing on its own is a far weaker signal than the mutation, which is worth keeping in mind when comparing them.
- **top50_focused_genes.csv:** the 50 strongest hits pulled out of Q1 and Q2 into one small table, so I could eyeball the biggest changes without opening the full files.
- **correlation_heatmap.png:** a square grid where every sample is compared with every other sample and the colour shows how similar they are. Samples from the same group should be more similar to each other than to other groups, so blocks of colour along the diagonal are what we want to see. It is a sanity check that the data is not scrambled and that no sample is wildly unlike its own group.
- **pca_top500.png, pca_allgenes.png, pca_Q1_genotype.png, pca_Q2_age.png:** PCA plots. PCA squashes thousands of genes down to two axes that capture the biggest sources of variation, so each dot is one fish and fish with similar expression sit close together. The four versions use different gene sets and colourings. These are what I used to check that the groups separate, and to spot the samples that sit away from their own group, which later led to the outlier sensitivity check.
- **volcano_q1_genotype.png, volcano_q2_age.png:** volcano plots. The x-axis is the log2 fold change (how much the gene moved, left is down and right is up) and the y-axis is the significance (higher is more confident). Genes in the top left and top right corners are the ones that moved a lot and are trustworthy. The Q1 volcano is heavily weighted to the left, which is the downregulation showing up visually.
- **venn_overlap.png:** a Venn diagram of the Q1 and Q2 DEG lists. The middle section is Q3, the 180 genes that are significant in both. What is interesting is not the number itself but that around 91 percent of those shared genes move in opposite directions in the two comparisons.
- **pvalue_distributions.png:** histograms of the raw p-values. This is a model diagnostic rather than a result. A healthy histogram is flat with a spike near zero. A strange shape would mean the statistical model is misbehaving, so this is a check before trusting anything else.
- **heatmap_top_degs.png:** a clustered heatmap of the strongest DEGs. Rows are genes, columns are samples, and the colour is expression. The clustering reorders both so that similar things sit together, which shows whether the samples naturally split by their real group.
- **ma_plots.png:** MA plots, which put average expression on the x-axis and fold change on the y-axis. They show whether the fold changes are behaving sensibly across the whole expression range rather than being an artefact of very low or very high counts.